

To address your question, I reviewed the document for information about threshold values for the Cranial Vault Asymmetry Index (CVAI) depending on whether the denominator is the shorter or longer diagonal diameter.

Reasoning and Approach:

- The document discusses the use of both the shorter diagonal (SD) and longer diagonal (LD) as denominators in the CVAI formula, noting that both are used in the literature and are mutually convertible.
- It specifically examines whether threshold values (diagnostic cutoffs) differ depending on which denominator is used.

Findings:

- The paper provides formulas to convert between CVAI(S) (shorter diagonal denominator) and CVAI(L) (longer diagonal denominator):
 - $CVAI(S) = CVAI(L) \times 100 / (100 - CVAI(L))$
 - $CVAI(L) = CVAI(S) \times 100 / (100 + CVAI(S))$
- The document explicitly addresses the threshold for deformational plagiocephaly:
 - "The diagnostic criteria threshold for deformational plagiocephaly, CVAI 3.5%, was converted to CVAI(L) 3.5% → CVAI(S) 3.6% and CVAI(S) 3.5% → CVAI(L) 3.4%, respectively."
- The differences between these threshold values are described as clinically negligible for the typical range (0%–15%).

Example Implementation:

- If a threshold of 3.5% is used with the longer diagonal as the denominator (CVAI(L)), the equivalent threshold using the shorter diagonal (CVAI(S)) would be 3.6%.
- Conversely, if 3.5% is used with the shorter diagonal, the equivalent for the longer diagonal would be 3.4%.

Summary Table:

Denominator Used	Threshold Value (%)
Shorter diagonal (SD)	3.5

Longer diagonal (LD)	3.4
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Or, if starting from LD 3.5%, SD would be 3.6%.

Conclusion:

- Yes, the paper does provide threshold values depending on whether the denominator is the shorter or longer diagonal diameter, and it offers formulas for converting between them. The differences are minor and considered clinically acceptable.

The reference:

Can you give the reference for this paper

Miyabayashi H, Saito K, Kato R, Noto T, Nagano N, Morioka I. Denominator of Cranial Vault Asymmetry Index: Choosing Between Longer and Shorter Diagonal Lengths. J Craniofac Surg. 2023;33(5):e369. doi:10.1097/SCS.00000000000009263